MILITARY SPECIFICATION

ACTUATOR, HYDRAULIC

This specification is approved for use within the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification defines the requirements of one type of hydraulic actuator controlled by a four-way electrohydraulic flow control servo valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-V-47308(MI)	Valve, Servo, Hydraulic Flow Control

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 1420
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MS3116	Connector, Plug, Electric, Series 1, Solder Type, Straight, Bayonet Coupling
MS33649	Bosses, Fluid Connection, Internal Straight Thread

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951	Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Materials. Materials and finishes to be used in construction of the actuator shall be selected in accordance with the order of precedence specified in MIL-STD-143.

3.3 Design and construction. The design and construction of the actuator shall be in accordance with this specification and figure 1 (see 4.6.1).

3.3.1 Weight. Weight of the actuator shall not exceed 4.0 pounds (see 4.6.1).

3.3.2 Fluid media. The actuator shall be designed to operate with hydraulic fluid conforming to MIL-H-5606 (see 4.6.1).

3.4 Performance.

3.4.1 Stroke. The actuator stroke shall be as shown on figure 1 (see 4.6.2).

3.4.2 Displacement polarity. A rated command to the servo valve causing flow out of valve port 1 shall extend the actuator (see 4.6.2).

3.4.3 Velocity. The actuator shall have a minimum zero load velocity of 4.3 inches per second for a maximum input flow of 1.8 gallons per minute (gpm) (see 4.6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

3.4.4 Input flow. The maximum required input flow with the servo valve commanded at rated input shall be 1.8 gpm for an actuator piston velocity of 4.3 inches per second (see 4.6.3).

3.4.5 Rod seal leakage. The total rod seal leakage shall not exceed four drops per hour under all operating conditions (see 4.6.3).

3.4.6 Proof pressure. The actuator shall withstand without evidence of internal or external leakage a proof pressure of 3020 pounds per square inch gage (psig) simultaneously applied to each chamber (see 4.6.4).

3.4.7 Stalled leakage. The actuator total stalled leakage shall not exceed 0.05 gpm for a 3000 psi differential (psid) pressure across the piston (see 4.6.4).

3.4.8 Stall force. The actuator shall have a stall force not less than 990 pounds with 675 psid across the piston and the maximum upstream pressure is 700 psig (see 4.6.5).

3.4.9 Breakaway differential pressure. The maximum differential pressure required to start piston motion in either direction shall not exceed 14 psid with zero external load (see 4.6.5).

3.4.10 External leakage. There shall be no evidence of external leakage, other than slight wetting insufficient to form a drop (see 4.6.6).

3.5 Environmental service conditions. The actuator shall operate satisfactorily, as specified, during or after exposure to the following environmental conditions (see 4.6.7).

3.5.1 Temperature.

3.5.1.1 Operating. The actuator shall operate as specified while being subjected to a temperature range of -31.7 to 71.1° Celsius (C) (-25 to 125° Fahrenheit (F)), (see 4.6.7.1.1).

3.5.1.2 Nonoperating. The actuator shall operate as specified after being subjected to a temperature range of -40 to 71.1°C (-40 to 160°F), (see 4.6.7.1.2).

3.5.2 Vibration. The actuator shall operate satisfactorily while being subjected to sinusoidal vibration at an intensity of 0.10 inch double amplitude (da) at a frequency of 5 to 10 Hertz (Hz), ± 0.51 gravity units (g) from 10 to 13 Hz, 0.060 inch da from 13 to 55 Hz and ± 10 g from 55 to 500 Hz (see 4.6.7.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

3.5.3 Shock. The actuator shall operate satisfactorily after being subjected to three 30 g half sine shocks of 11 milliseconds (ms) duration in each of the three orthogonal axes (see 4.6.7.3).

3.6 Marking. The valve shall be marked in accordance with the requirements of MIL-STD-130 (see 4.6.1).

3.7 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9 (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the conditions of 4.3.1, 4.3.2, 4.3.3 and 4.3.4.

4.3.1 Temperature. Temperature shall be room ambient, 26.6 ± 4°C (80 ± 7°F).

4.3.2 Barometric pressure. Barometric pressure shall be facility ambient, 28 to 32 inches of mercury (in Hg).

4.3.3 Humidity. Humidity shall be up to 95 percent relative humidity.

4.3.4 Fluid temperature and pressure. Fluid temperature shall be 37.8 ± 11.1°C (100 ± 20°F). Supply pressure shall be 1000 + 200, -300 psig with a return pressure of 20 ± 5 psig.

4.3.5 Inspection equipment. Special inspection equipment for conducting inspections and tests shall consist of a test manifold to simulate ports 1 and 2 of the flow control valve conforming to MIL-V-47308(MI) and a complete flow control valve conforming to MIL-V-47308(MI).

4.4 First article inspection. First article inspection shall consist of the inspection and tests specified in table I. The sequence of inspection shall be at the option of the contractor. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity (see 3.1 and 6.2).

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Temperature	3.5.1	4.6.7.1
Operating	3.5.1.1	4.6.7.1.1
Nonoperating	3.5.1.2	4.6.7.1.2
Vibration	3.5.2	4.6.7.2
Shock	3.5.3	4.6.7.3

4.4.1 First article sample. The first article sample shall consist of four units, representative of the production product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection. Quality conformance inspection shall be conducted on the basis of single sampling of a random sample selected from each inspection lot or less for an acceptable quality level of 1.0 percent, in accordance with MIL-STD-105 (see 6.2).

4.5.1 Inspection lot. The lot size shall not exceed the quantity of items produced in one continuous operation utilizing the same production processes for delivery for one contract or order.

4.5.2 Inspection and tests. Samples selected in accordance with 4.5 shall be subjected to the inspections and tests in table II. Sequence of inspections shall be at the option of the contractor.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product	3.3, 3.3.1, 3.3.2, 3.6 and 3.7	4.6.1
Stroke	3.4.1	4.6.2
Displacement polarity	3.4.2	4.6.2
Velocity	3.4.3	4.6.3
Input flow	3.4.4	4.6.3
Rod seal leakage	3.4.5	4.6.3
Proof pressure	3.4.6	4.6.4
Stalled leakage	3.4.7	4.6.4
Stall force	3.4.8	4.6.5
Breakaway differential pressure	3.4.9	4.6.5
External leakage	3.4.10	4.6.7

4.5.3 Noncompliance. If any unit fails to pass inspection, units may be reworked to correct defects and resubmitted for inspection. However, before reinspection, the manufacturer shall notify the procuring activity and the cognizant inspection activity of such failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the procuring activity has been taken. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the procuring activity.

4.5.4 Inspection of packaging. Preservation, packing and marking shall be inspected for conformance to section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

4.6 Methods of inspection.

4.6.1 Examination of product. Each valve shall be visually and dimensionally inspected and weighed to determine conformance to the requirements of this specification (see 3.3, 3.3.1, 3.3.2, 3.6 and 3.7).

4.6.2 Stroke and polarity. With the flow control valve of 4.3.5 installed on the actuator and the pressures of 4.3.4 applied, the valve shall be tested for conformance to 3.4.1 and 3.4.2. The actuator shall be inspected for conformance to 3.4.10.

4.6.3 Velocity, input flow and rod seal leakage. With the flow control valve of 4.3.5 installed on the actuator and the actuator in the test configuration shown on figure 2, the pressures of 4.3.4 shall be applied and the actuator tested for conformance to 3.4.3, 3.4.4 and 3.4.5. The actuator shall be inspected for conformance to 3.4.10.

4.6.4 Proof pressure and stalled leakage. With the test manifold of 4.3.5 installed and the actuator piston locked at midstroke, the actuator shall be proof tested for conformance to 3.4.6. The actuator shall be tested for stalled leakage at midstroke and at positions ±0.75 inches from midstroke for conformance to 3.4.7. The actuator shall be inspected for conformance to 3.4.10.

4.6.5 Stalled force and breakaway differential pressure. With the test manifold of 4.3.5 installed and the actuator loaded, the actuator shall be tested extended, 25 percent retracted, 50 percent retracted, 75 percent retracted, fully retracted, 25 percent extended, 50 percent extended and 75 percent extended for conformance to 3.4.8. The above will be repeated with the actuator unloaded to determine conformance to 3.4.9. The actuator shall be inspected for conformance to 3.4.10.

4.6.6 External leakage. The actuator shall be inspected for conformance to 3.4.10 after each test.

4.6.7 Environmental service conditions. The actuator shall be tested for conformance to 3.5.

4.6.7.1 Temperature.

4.6.7.1.1 Operating. The actuator shall be tested in accordance with MIL-STD-810, method 501, procedure II to determine conformance to 3.5.1.1 for high temperature. The actuator shall be tested in accordance with MIL-STD-810, method 502, procedure II to determine conformance to 3.5.1.1 for low temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

4.6.7.1.2 Nonoperating. The actuator shall be tested in accordance with MIL-STD-810, method 501, procedure I to determine conformance to 3.5.1.2 for high temperature. The actuator shall be tested in accordance with MIL-STD-810, method 502, procedure I to determine conformance to 3.5.1.2 for low temperature.

4.6.7.2 Vibration. The actuator shall be tested in accordance with MIL-STD-810, category 5 to determine conformance to 3.5.2.

4.6.7.3 Shock. The actuator shall be tested in accordance with MIL-STD-810, procedure I to determine conformance to 3.5.3.

5. PACKAGING

5.1 Preservation. Preservation shall be industrial in accordance with ASTM D3951.

5.2 Packing. Packing shall be industrial in accordance with ASTM D3951.

5.3 Marking. In addition to any special or other identification marking required by the contract or purchase order (see 6.2) each unit pack and exterior container shall be marked in accordance with ASTM D3951.

6. NOTES

6.1 Intended use. This actuator is intended for use in a military subsystem.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Whether first article sample is required (see 3.1 and 4.4).

c. Sampling plan if other than specified (see 4.5).

d. Applicable level of preservation, packing and marking if other than specified (see section 5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of four units. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

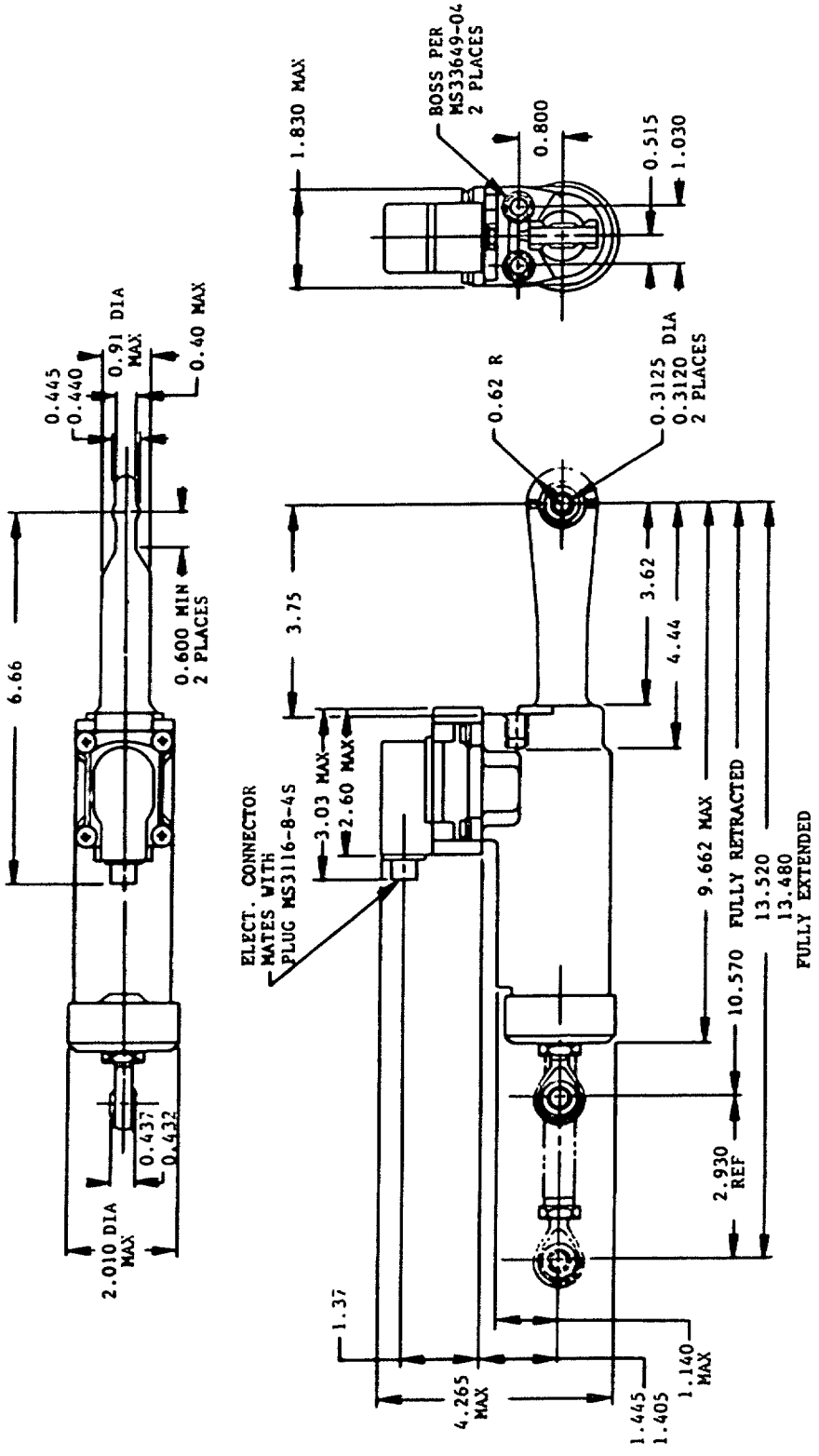
Actuator, hydraulic

Custodian:
Army - MI

Preparing activity:
Army - MI

(Project No. 1420-A288)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.



NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 1. Actuator, hydraulic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

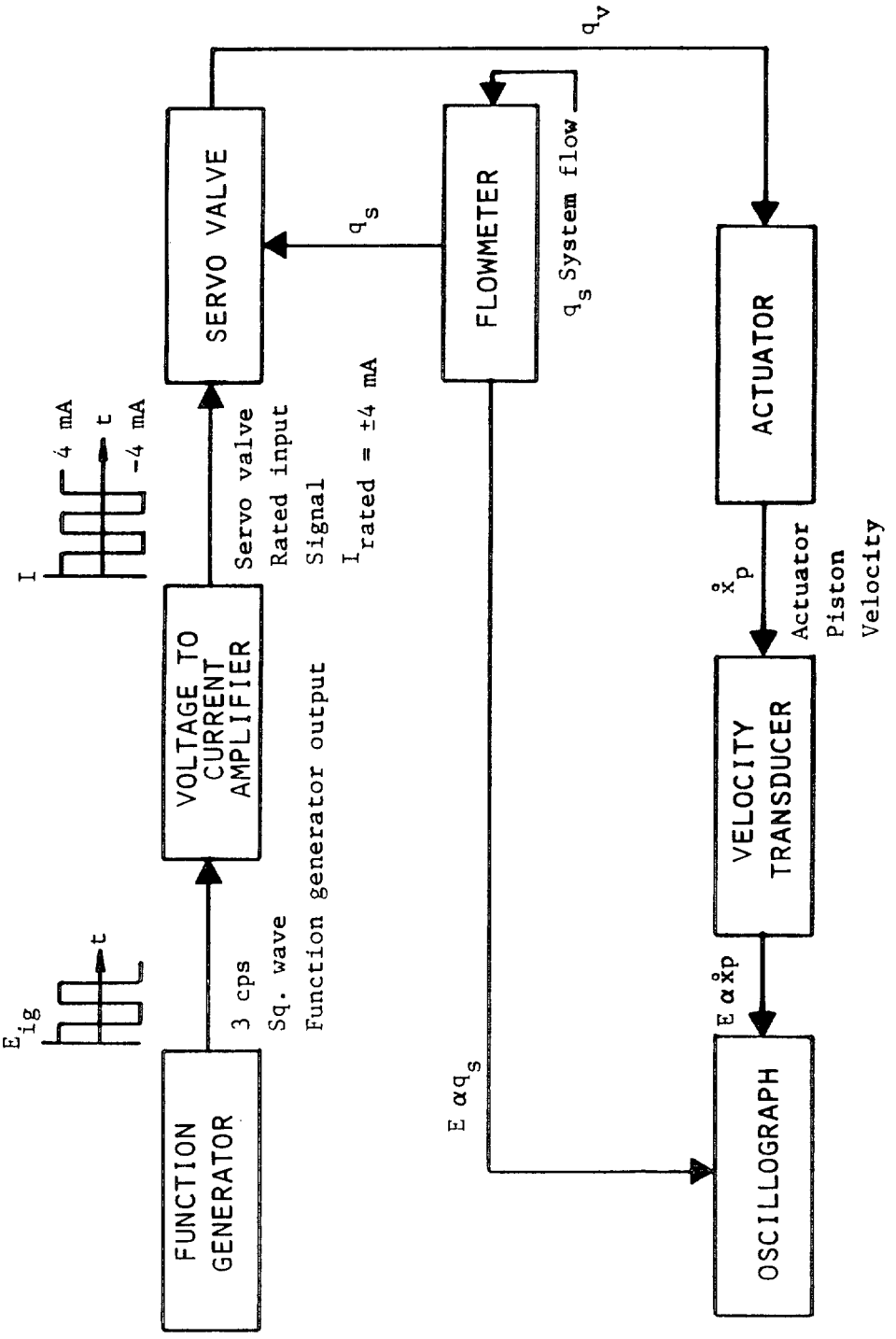


FIGURE 2. Actuator velocity testing, block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-47312(MI)	Actuator, Hydraulic

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.